Karthikay Chandana

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EDUCATION

Institution	Degree / Board	Passing Year	Score / CGPA
Indian Institute of Technology Hyderabad	B.Tech	2028	8.87
Narayana E-Techno School Mumbai	Class XII (Central Board of Secondary Education)	2024	96.20%
R N Podar High School Mumbai	Class X (Central Board of Secondary Education)	2022	95.00%

SCHOLASTIC ACHIEVEMENTS

- Finished **17th in Optiver Trade-A-Thon** held on 23rd March 2026 among 400+ participants
- **India Rank 3** and **Worldwide Rank 36** amongst 20000+ teams after the penultimate round of IMC Prosperity 4
- **Rank 620** in IICPC Quantfest 2025 Prelims among 10000+ participants
- NSEA: Among **top 1%** performers in Maharashtra in NSEA 2023
- National **top 1%** in IOQJS 2021, part 1. Qualified for part 2
- TIMO Gold Medalist, IJMO Silver Medalist, SASMO Silver Medalist and SEAMO Bronze Medalist in 2021

PROJECTS

Ebullient Securities PS | *Jupyter Notebooks, Git*

Inter IIT Techmeet 14.0

- Collaborated within a 10-member team to develop a quantitative trading strategy for the competition problem statement.
- Explored and evaluated multiple approaches, including **Fourier Transform** based methods and **custom probabilistic models**, before converging on a **modified VIDYA** (Variable Index Dynamic Average) indicator complemented by **statistically driven** trading strategies.
- Achieved a Sharpe ratio of **2.61** and a Calmar ratio of **2.7** on the provided dataset.

5-Stage Pipelined RISC-V Simulator | *C++, Python, Git* | *GitHub Link*

Computer Architecture Course

- Extended the in-house RISC-V simulator to support **multiple pipelined execution modes**, including hazard detection, full forwarding logic, and both static and N-bit dynamic branch prediction.
- Implemented a **highly configurable read/write cache** system supporting **LRU**, **Write-Allocate**, and **Write-Back** policies, alongside advanced functionalities like **undo/redo commands** and performance tracking metrics.
- Built a python based analysis tool using **Matplotlib** and **Pandas** to visualize execution latency across pipeline stages, and comprehensively verified the pipelined design against a single-cycle baseline.

xv6 Kernel Extensions | *C, xv6 OS* | *GitHub Link*

Operating Systems Course

- Extended the xv6 kernel with **custom system calls** and a **4-level-MLFQ scheduler** with priority boosting, preemption, and scheduling analytics.

- Developed a **clock-based page replacement algorithm** to manage physical frames, handle page faults, and support disk-backed virtual memory.
- Implemented **FCFS** and **SSTF** disk scheduling algorithms along with **RAID-0, RAID-1,** and **RAID-5** storage support for fault tolerance and performance.

Infinite Precision Calculator | *Java, Git* | *GitHub Link*

Software Development Fundamentals Course

- Developed a **custom Java library** featuring AInteger and AFloat classes to bypass native primitive data type limitations, storing values as strings to **execute arithmetic operations** digit-wise on **infinite-length numbers**.
- Programmed **fundamental algorithms**, including classical long multiplication, manual long division, and decimal alignment mechanisms, supporting floating-point computations up to **30 digits of precision** with sign handling.

RELEVANT COURSES

- **Mathematics:** Discrete Maths, Linear Algebra, Probability and Random Variables, Calculus, Introduction to Statistics
- **Computer Science:** Data Structures and Algorithms, Computer Architecture, Software Development Fundamentals, Database and Management Systems, Operating Systems, Theory of Computation, Introduction to Programming

- **Additional Learning:** Completed CS50X, CS50P, and CS50AI (Harvard University, via edX). Earned certification in AP Microeconomics (MIT, via edX).

SKILLS

- **Programming Languages:** C, C++, Python, SQL, Shell Script, Java
- **Libraries:** NumPy, Pandas, Matplotlib
- **Tools:** Git, Jupyter Notebooks, Linux, LaTeX

POSITIONS OF RESPONSIBILITY

- **Internship Coordinator, OCS** (Office of Career Services): **Head Internship Coordinator** for the CSE Batch, assisting in pitching and coordinating with companies for recruitment

- **Core, Infero** (Competitive Programming Club): Assisted with **problem setting** for several competitions

EXTRACURRICULAR

- Chess: Hold an international **FIDE rating of 1692**. Represented Karnataka in Nationals and competed in **100+ tournaments**, in India and abroad.

- Actively **track** and **invest** in the Indian equity markets, managing a personal portfolio that has generated an **XIRR of 7.07%** over the past **2 years**

- Competitive Programming: Engaged in competitive programming, peak rating **1458** (Sanctimonious) on codeforces

- CTF: Participated in multiple CTFs. Won gold in MILAN **Inter-Hostel** CTF competition

- Volunteer at Parikrma Foundation Bangalore. Taught multiple **class 6th and 7th** students maths **over 3 weeks**